Camouflage Unlearning: Fast and Surgical Machine Unlearning for Biometric Systems

***Abstract*—Machine unlearning for privacy-critical biometric systems, fingerprint, iris, and face recognition, remains underexplored despite GDPR/CCPA mandates requiring erasure throughout deployment. Existing methods target classification and generative tasks, collapsing all forget classes to a single distant point, which inflates convergence time on resource-constrained edge devices facing frequent, minimal-replay removal requests; they are also benchmarked almost exclusively on ResNet/ViT for CIFAR/ImageNet, leaving biometric backbones untested. We propose Camouflage Unlearning (CU), which maps forget classes to nearby retain-class boundaries instead of a single point, reducing migration distance; this proximity target shapes only the training-time optimization objective, and we verify empirically that it does not induce elevated impostor matches against the assigned retain identity at verification time. CU achieves 3–4× faster convergence via retain-dominance constraints and localized regularization that surgically updates specific transformer blocks, while maintaining competitive retain accuracy. Across classification, iris, fingerprint, and face recognition, CU shows competitive performance and privacy behavior (MIA ≈ 0.5), supporting its effectiveness for regulatory-compliant, deep embedding-based biometric authentication.**

***Index Terms*—Biometric recognition, data privacy, face recognition, GDPR, machine unlearning, vision transformers**

I. Introduction

Machine unlearning (MU) has emerged as a critical research area, addressing the imperative to prevent unauthorized data exploitation, eliminate harmful model behaviors, and safeguard user privacy [1]. The proliferation of AI misuse ranging from phishing generation and model inversion attacks to perpetuating misinformation has prompted regulatory frameworks [2], [3], [4]. Frameworks like the General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) [5], [6] mandate that model service providers erase user data from learned representations upon request, establishing legal and ethical obligations for responsible AI deployment. Users possess the right to request data removal at any time, so per-request efficiency becomes a first-class concern.

Existing machine unlearning methods predominantly target image classification [7] and generative tasks [8], with limited exploration of privacy-critical biometric systems. While face recognition [9], [10], [11] has received modest attention, fingerprint and iris recognition remains substantially less studied. This gap persists despite their sensitive nature and widespread deployment in authentication systems. Contemporary approaches rely primarily on KL divergence maximization [12] or gradient ascent [13]. These methods implement many-to-one mappings that force forgotten class embeddings toward a single distant point in feature space. Alternative strategies include saliency-based selection [14], geometric boundary manipulation [15], perturbation-based forgetting [16], and meta-learning optimization [17], [18]. A key observation across all these methods is that every forget sample is driven toward a single collapse point in the decision space. Even when a sample already lies far from the retain manifold, it must still traverse the full distance to this distant target. This increases convergence time and GPU cost, as observed in Table I. This many-to-one collapse behavior is visualized and ablated in detail in Section III-B. Concretely, GS-LoRA [13] (group-sparse LoRA, slow ascent), SCRUB [12] (bidirectional logits, unstable KL), SalUn [14] (weight saliency, reversible labels), Boundary Unlearning [15] (shadow-node pruning, slow), Forget Vectors [16] (noise tuning, invertible), MCU [18] (Bézier curves, dual optimization), LTU [17] (meta-learning, computationally expensive), DELETE [19] and LoTUS [20] (frozen teacher, double memory), and MUNBa [21] (Nash bargaining, per-step overhead) each trade convergence speed, memory, or reversibility, and few explicitly target biometric edge deployment or sequential forget cycles.

Biometric recognition differs from general classification in ways that directly affect unlearning feasibility. Biometric systems deploy on resource-constrained edge devices. Removal requests arrive sequentially and frequently over the deployment lifecycle. This transforms unlearning into a recurring operational cost rather than a one-time procedure. Under this workload, existing many-to-one methods become prohibitive as per request compute scales poorly with request frequency. Compounding this, existing unlearning methods have been designed and benchmarked almost exclusively on ResNet [22] or ViT backbones [23] for CIFAR [24] and ImageNet [25]. Their transferability to biometric backbones such as Face Transformer [26] (sliding-patch ViT face recognition), LAM [27] (ResNet-50, lightweight attention), JIPNet [28] (CNN-Transformer joint pose-identity), SSLDMNet-Iris [29] (CNN-GRU, Fisher discriminant), IrisFormer [30] (rotation-invariant ViT matching), PFSimNet [31] (partial fingerprint similarity network), and MEM-FR [32] (optical, privacy-preserving recognition) remains untested, despite these architectures representing the state of the art among deep embedding-based biometric backbones. What biometric unlearning requires instead is a cycle-robust method validated directly on biometric backbones.

To address these challenges, we propose Camouflage Unlearning (CU), a novel framework that achieves 3–4× faster convergence while maintaining robust performance. Unlike existing many-to-one approaches, CU pushes forget-class embeddings toward the nearest retained class boundaries, effectively reducing migration distances. This proximity target governs only the training-time optimization objective, not final verification-level similarity; a naive reading might suggest CU induces impersonation of the nearest retain identity, but

Section III-D empirically verifies that forget probes do not match their assigned retain centroid more often than chance, ruling out this concern. To perform surgical modifications and minimize collateral damage to retain classes, CU employs localized regularization concentrated in specific blocks of the neural network for additional details, please refer to Section II.

Our main contributions are summarized as follows:

1) To the best of our knowledge, this is among the earliest studies of machine unlearning for privacy-critical deep embedding-based biometric systems, including fingerprint, iris, and face recognition.
2) We introduce Camouflage Unlearning, which achieves 3–4× faster convergence compared to SoTA unlearning methods while maintaining competitive retention accuracy and privacy guarantees.
3) We conduct extensive empirical evaluation across multiple biometric and classification domains, demonstrating that CU concentrates updates in specific network blocks and remains effective under data-scarce replay conditions.

II. Problem Setting and Methodology

A. Problem Setting

Let $\mathcal{M}$ be a model pre-trained on $D = D_r^{full} \cup D_f$ ($D_r^{full} \cap D_f = \emptyset$), mapping $f_{\mathcal{M}} : \mathcal{X}_D \to \mathcal{Y}_D$. Since storing D_r^{full} is prohibited under GDPR/CCPA and full retraining is costly, we retain only a small replay buffer $D_r \subset D_r^{full}$, at most 10% of $|D_r^{full}|$[1]. Before unlearning, $\mathcal{M}$ correctly maps D_f and D_r; applying $\mathcal{F}$ (Section II-B) yields $\mathcal{M}' = \mathcal{F}(\mathcal{M}, D_f, D_r)$, under which the forget mapping should no longer hold while the retain mapping persists:

$$f_{\mathcal{M}} : \mathcal{X}_{D_f} \to \mathcal{Y}_{D_f}\,;\ \mathcal{X}_{D_r} \to \mathcal{Y}_{D_r} \implies f_{\mathcal{M}'} : \mathcal{X}_{D_f} \not\to \mathcal{Y}_{D_f}\,;\ \mathcal{X}_{D_r} \to \mathcal{Y}_{D_r} \tag{1}$$

where $\not\to$/$\to$ denote forgotten/retained mappings. Successful unlearning requires:

- Forgetting: $Acc(\mathcal{M}', D_f) \leq \frac{1}{C}$,
- Retention: $Acc(\mathcal{M}', D_r) \approx Acc(Oracle, D_r)$

where C is the number of classes and Oracle denotes a model trained directly on target classes only.

B. Camouflage Unlearning

CU relocates forget-class embeddings to retain-class boundaries via direct centroid mapping, a many-to-many distribution replacing the many-to-one collapse of existing methods, reducing migration distance and accelerating convergence.

We compute class centroids by averaging embeddings, $c_k = \frac{1}{|D_k|}\sum_{x \in D_k} emb(x)$, yielding retain centroids $\{c_{r_1}, \ldots, c_{r_K}\}$ and forget centroids $\{c_{f_1}, \ldots, c_{f_M}\}$. Each forget centroid is assigned its nearest retain centroid as target:

$$\mathbf{t}_i = c_{r_{j^*}}, \quad j^* = \arg\min_j \|c_{f_i} - c_{r_j}\|_2 \tag{2}$$

We pull forget embeddings toward their assigned targets:

$$\mathcal{L}_f = \frac{1}{|D_f|} \sum_{(\mathbf{x},y) \in D_f} \|emb(\mathbf{x}) - \mathbf{t}_y\|^2 \tag{3}$$

[1] From this point, D_r denotes the replay buffer unless specified.

This risks corrupting retain decision regions and prolonging convergence, so we enforce that retain logits dominate forget logits by margin $\gamma = 2.0$:

$$\mathcal{L}_{dom} = ReLU\left(\max_{i \in F} z_i + \gamma - \max_{j \in R} z_j\right) \tag{4}$$

where z_i are logits and F, R index forget/retain classes; this accelerates forget suppression while preserving retain accuracy. Retain knowledge is further preserved via classification and embedding stability:

$$\mathcal{L}_{ret} = \lambda_{ce} \cdot CE(\mathbf{z}_r, y_r) + \lambda_{emb} \cdot \frac{1}{|D_r|} \sum_{(\mathbf{x},y) \in D_r} \|emb(\mathbf{x}) - c_y\|^2 \tag{5}$$

where $\mathbf{z}_r$, y_r, and c_y are retain logits, labels, and original centroids. Since full-parameter updates risk instability under limited replay, inspired by group sparse selection [13], we apply localized regularization over fewer transformer blocks:

$$\mathcal{L}_{local} = \lambda_{local} \sum_{g=1}^{G} \|\theta_g\|_2 \tag{6}$$

where θ_g are block-g parameters and G is the number of groups, each initialized at zero to track the *change* from pretrained weights, so $\mathcal{L}_{local}$ penalizes update magnitude rather than absolute weights. This group-Lasso approach concentrates updates in few blocks, preserving retained knowledge under constrained replay. The total loss:

$$\mathcal{L}_{total} = \mathcal{L}_f + \lambda_{dom}\mathcal{L}_{dom} + \mathcal{L}_{ret} + \mathcal{L}_{local} \tag{7}$$

III. Performance Evaluation

A. Experimental Setup

Datasets, Backbones, and Baselines: We evaluate on CIFAR-100 [24] (DeiT-Tiny [33]), CASIA-Iris [34] (LAM [27]), Sokoto Coventry fingerprints [35] (JIPNet [28]), and CASIA-Face100 [36] (Face Transformer [26]), comparing CU against SCRUB [12], SalUn [14], GS-LoRA [13], LTU [17], MCU [18], and Forget Vectors [16], plus an Oracle trained only on $D_r^{\mathbf{full}}$ as the upper bound.

Evaluation Metrics: We report Acc_f, Acc_r, MIA (label-only attack [37], ≈ 0.5 = safe), KL divergence to Oracle, RTE, and H-Mean [38]. Success requires $Acc_f \approx 0$, $Acc_r \approx$ Oracle, low KL/RTE. FAR@$\{0.3, 0.4, 0.5\}$/EER (cosine similarity, forget probes vs. retain gallery) further verify open-set identity confusion (Section III-D).

B. Main Results

Table I compares all methods across four tasks; green/blue mark best/second-best.

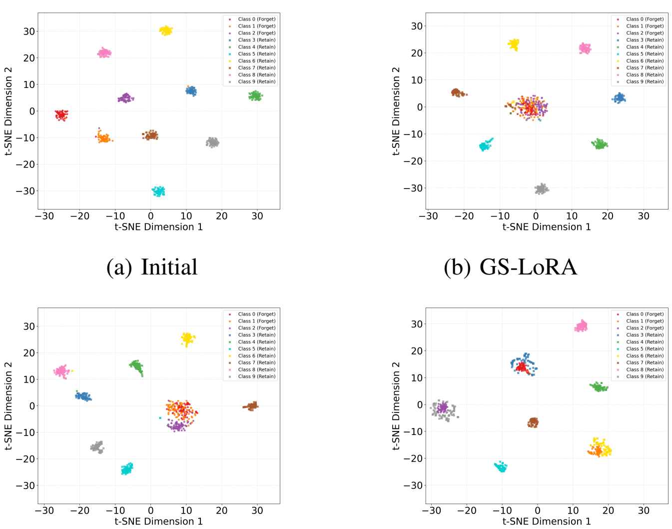
Results: All methods forget near-perfectly ($Acc_f < 1\%$). CU ranks first or second in Acc_r and H-Mean across all tasks (LTU and MCU the closest competitors), and achieves MIA ≈ 0.5 despite GS-LoRA's lower KL, expected since CU deliberately reshapes forget logits. Most critically, CU delivers a 3–4× RTE speedup over every baseline.

Method	Classification						Iris recognition					
	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓
Oracle	–	77.36	–	–	–	–	–	99.03	–	–	–	–
GS-LoRA [13]	0.11	71.32	74.18	0.73	0.56	153.2	0.00	93.42	96.18	0.75	0.69	225.7
SalUn [14]	0.37	72.46	74.63	1.32	0.63	225.6	0.37	92.37	95.44	0.79	0.59	427.2
SCRUB [12]	0.58	73.11	74.90	1.89	0.64	219.3	0.43	95.58	97.07	0.83	0.63	501.8
FV [16]	0.00	74.43	75.89	0.87	0.55	189.5	0.00	96.27	97.63	1.13	0.57	383.1
MCU [18]	0.42	74.43	75.66	0.87	0.59	175.5	0.00	98.87	98.95	0.99	0.63	437.2
LTU [17]	0.00	76.32	76.83	0.75	0.57	239.3	0.73	95.43	96.85	1.25	0.55	329.4
CU	0.47	74.46	75.65	1.27	0.56	55.5	0.06	97.89	98.43	1.01	0.53	116.1

Method	Fingerprint recognition						Face recognition					
	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓	Acc_f↓	Acc_r↑	H-$Mean$↑	KL↓	MIA↓	RTE↓
Oracle	–	90.82	–	–	–	–	–	95.12	–	–	–	–
GS-LoRA [13]	0.00	83.42	86.98	1.13	0.65	155.3	0.57	93.25	93.90	1.13	0.53	284.4
SalUn [14]	0.10	82.32	86.24	0.75	0.63	202.2	0.00	90.15	92.56	1.27	0.59	417.2
SCRUB [12]	0.00	83.27	86.85	2.36	0.57	175.3	0.42	93.27	93.98	2.87	0.63	389.5
FV [16]	0.30	84.44	87.38	1.27	0.54	160.7	0.20	92.27	93.58	1.20	0.60	380.3
MCU [18]	0.00	80.32	85.24	0.99	0.59	220.5	0.00	90.12	92.54	0.89	0.57	403.8
LTU [17]	0.00	87.72	89.24	1.13	0.54	237.4	0.00	92.58	93.84	0.93	0.55	430.2
CU	0.00	87.23	88.98	1.05	0.55	55.32	0.00	93.98	94.55	1.67	0.51	175.3

TABLE I
PERFORMANCE ACROSS FOUR DOMAINS; CU ACHIEVES 3–4× RTE SPEEDUP.

Geometric Analysis: Figure 1 visualizes embedding-space t-SNE. GS-LoRA [13], SCRUB [12], and Boundary Unlearning [15] collapse forget classes to one distant point; CU instead migrates them to the nearest retain boundary, explaining its RTE gains in Table I.



(a) Initial (b) GS-LoRA (c) Boundary (d) CU (Ours)

Fig. 1. t-SNE: CU spreads embeddings; baselines collapse to one point.

C. Ablation Studies

Replay Buffer Size: We investigate the minimal replay buffer size $|D_\mathbf{r}|$ for effective unlearning on CIFAR-100. Table II shows that a 0.1 ratio optimally balances retention accuracy and efficiency.

Buffer Ratio	Speed	Acc_f	Acc_r	H-$Mean$
1.0	1×	1.33	74.39	2.62
0.5	2×	1.18	73.00	2.32
0.3	3.3×	1.33	74.55	2.61
0.1	10×	1.30	74.52	2.56

TABLE II
REPLAY BUFFER SIZE ABLATION; 0.1 RATIO BALANCES RETENTION/EFFICIENCY.

Retain Dominance Constraint: Without this constraint, we found that the forget accuracy exhibits a prolonged tail, delaying convergence. Table III demonstrates that applying the constraint reduces Acc_f from 9.97% to 0.39%, achieving faster unlearning with minimal retention loss.

Method	Acc_f	Acc_r
Without Constraint	9.97	78.12
With Constraint	0.39	77.06

TABLE III
DOMINANCE CONSTRAINT ELIMINATES LONG-TAIL CONVERGENCE.

Localized Parameter Regularization: This regularization selectively modifies fewer blocks, enabling surgical edits that preserve model capacity. On CIFAR-100 classification, Figure 2 shows concentrated updates in specific blocks, while Table IV demonstrates reducing total ℓ_2 norm from 39.05 to 10.09 with minimal retention loss.

Method	Acc_f	Acc_r	Total ℓ_2 Norm
Without Localized Reg	0.00	77.24	39.05
With Localized Reg	0.00	74.76	10.09

TABLE IV
LOCALIZED REGULARIZATION LOWERS ℓ_2 NORM; MINIMAL ACCURACY LOSS.

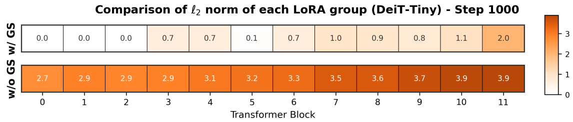


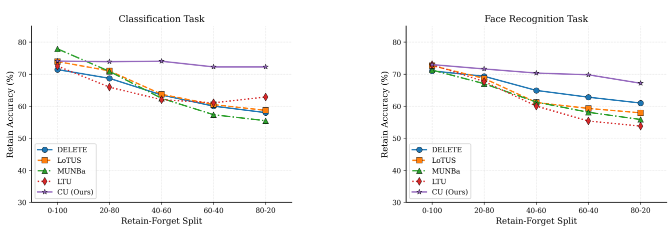
Fig. 2. Per-block ℓ_2 norm: updates concentrate in few blocks.

Scalability: We apply CU to five ViT backbones (DeiT-Tiny to ViT-Large). Table V shows consistent forgetting and retention across both lightweight and large-scale architectures.

Model	Before Unlearning		After Unlearning	
	Acc_f	Acc_r	$Acc_f \downarrow$	$Acc_r \uparrow$
DeiT-Tiny	75.25	75.67	0.40	74.46
DeiT-Small	74.86	74.01	1.35	73.89
DeiT-Base	74.32	74.15	0.87	72.58
ViT-Base	72.58	73.21	2.35	70.35
ViT-Large	75.35	78.91	0.10	75.61

TABLE V
VIT BACKBONE SCALE VS. UNLEARNING PERFORMANCE.

Sequential Removal Workload: We pilot CU against DELETE [19], LoTUS [20], MUNBa [21], and LTU [17] across successive forget cycles on CIFAR-100 [24] and CASIA-Face100 [36]. Figure 3 shows baselines degrade progressively while CU retains stable accuracy, confirming robustness under continual removal.



(a) CIFAR-100 (b) Face Recognition

Fig. 3. Sequential-cycle retain accuracy; baselines degrade, CU stable.

D. Attack-Based Verification

Table VI reports FAR/EER of forget probes vs. the retain gallery on CASIA-Iris. CU tracks the Oracle closely (FAR@0.5 3.83% vs. 3.10%; EER 21.99% vs. 21.69%), both rising similarly over *Before* (1.64%), so the small increase stems from removing identities, not from CU.

Model	FAR@0.3	FAR@0.4	FAR@0.5	EER
Before	11.38%	4.84%	1.64%	13.33%
CU	17.97%	8.52%	3.83%	21.99%
Oracle	16.10%	7.79%	3.10%	21.69%

TABLE VI
FAR/EER OF FORGET PROBES VS. RETAIN GALLERY (CASIA-IRIS).

Aggregate FAR/EER cannot rule out a concentrated, backdoor-like confound between one forget identity and its specifically assigned nearest-retain centroid, since such a pairing would barely move the aggregate rate. Table VII reports the discriminating per-identity measurement directly, each forget probe's match rate against its *assigned* retain centroid, alongside linear-probe and relearning tests for backbone-level forgetting and further MIA/inversion attacks, on CASIA-Face100 after all forget identities are removed, referenced against the Oracle. The assigned-centroid match rate is no higher for CU than for Oracle, ruling out concentrated confounding between individual forget–retain pairs; the linear-probe rate (within a point of Oracle) and slow relearning show forgetting is diffuse and not easily reversed, and label-only/adaptive MIA near chance with a low inversion match rate show stronger, adaptive attacks also fail.

Test	CU	Oracle
Assigned-centroid match $\downarrow$	**2.71**	3.54
Linear probe (frozen emb.) $\downarrow$	**5.28**	4.85
5-shot relearn $Acc_f \downarrow$	**4.37**	7.12
MIA, label-only [37] ≈ 0.5	**0.52**	0.51
MIA, adaptive ≈ 0.5	**0.54**	0.54
Model inversion match $\downarrow$	**3.42**	3.78

TABLE VII
PER-IDENTITY CONFOUND TESTS ON CASIA-FACE100, ANSWERING WHETHER A FORGET PROBE MATCHES ITS SPECIFICALLY ASSIGNED RETAIN CENTROID MORE THAN CHANCE.

Attention Maps: We visualize attention on forget/retain CASIA-Face100 [36] samples before/after CU. Figure 4 shows the base model attends to facial regions for both; retrained and CU models shift attention to peripheral areas *only for forget samples*, confirming CU surgically removes identity-specific features while preserving retain ones.

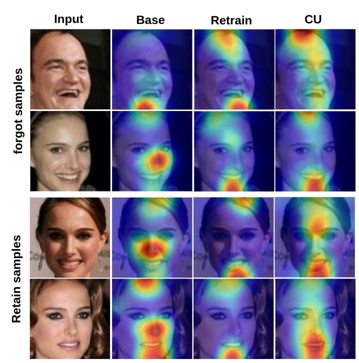


Fig. 4. Attention maps: retrained/CU models disperse attention off-face only for forget identities.

Relearning Attacks: We test whether CU's forgetting is genuine or reversible via a relearning attack: fine-tuning a CU-unlearned model (50 classes) and a head-masked baseline on forgotten classes, tracking recovery. Figure 5 shows CU recovers gradually versus head-masking's rapid recovery, confirming genuine backbone-level forgetting.

IV. RESULTS DISCUSSION

A. On the Necessity of the Retain Buffer

A natural concern is whether the replay buffer $D_r \subset D_r^{\text{full}}$ contradicts GDPR/CCPA erasure mandates. We argue D_r

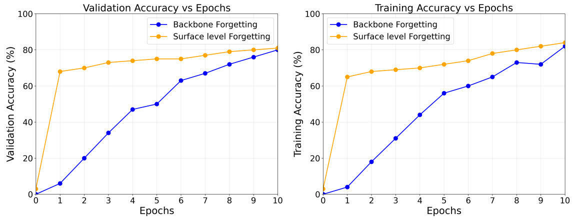


Fig. 5. CU recovers gradually (genuine) vs. rapid head-masking (superficial).

is unavoidable: pretraining entangles D_f and D_r^{full} in the same parameters, so disentangling them needs a retained reference absent architectural support (e.g., modular pathways). Existing alternatives fall short: impair-and-repair [39] still needs D_r for repair, source-free unlearning [40] works only for convex/linear models, and all non-convex methods, via Fisher/saliency [41], [14], retention losses [12], [13], or continual mechanisms [38], [42], [43], [44], [45], rely on D_r; only generative-task exceptions FLAT [46] and ESD [47] avoid it. CU keeps a minimal D_r, at most 10% of D_r^{full}, at least one sample per retain identity, per GDPR/CCPA. Retain metrics use the full held-out split, not D_r. D_r holds only *consenting* identities: withdrawal is treated as a forget request, and the buffer refills from the consenting pool.

B. Scope Across Modalities

While face recognition is dominated by deep embedding models, several deployed iris and fingerprint systems still rely on classical, non-learned representations (e.g., Daugman iris codes, minutiae matching). The reported gains of CU apply to deep embedding-based biometric backbones; extending camouflage unlearning to classical, non-differentiable feature pipelines is outside the current scope.

V. Conclusion

We are among the earliest to address machine unlearning for privacy-critical deep embedding-based biometric systems, fingerprint, iris, and face recognition. Existing methods suffer from prolonged convergence due to many-to-one embedding collapse, inefficient for biometric models facing frequent unlearning requests on edge devices. We propose Camouflage Unlearning, achieving up to $4\times$ speedup via many-to-many distribution and surgical parameter modifications, with forgetting, retention, and privacy performance comparable to state-of-the-art methods. Future work extends CU to continual forgetting in pretrained biometric models.

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